A Lightweight Credit Signal Decorator via Group-Relative Counterfactual Advantage for Cooperative Multi-Agent Reinforcement Learning

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Abstract. Reliable individual credit assignment remains difficult in cooperative multi-agent reinforcement learning under *Centralized Training with Decentralized Execution* (CTDE), where agents act from local observations while learning from shared team rewards. We propose **GrCA-Decor**, a lightweight add-on for value-based CTDE methods that augments each agent utility with a peer-relative signal called *Group-Relative Counterfactual Advantage* (GrCA). During training, GrCA subtracts a same-action group-mean value from the chosen action value of each agent and feeds the resulting relative correction into the per-agent utility before mixing. This design provides an auxiliary utility-shaping signal while preserving the decentralized execution interface and avoiding a heavier interaction model. Experiments on the *StarCraft Multi-Agent Challenge* (SMAC) and *Google Research Football* benchmarks show that GrCA-Decor improves representative value-based baselines, with the clearest gains appearing on harder coordination tasks. These results support peer-relative utility shaping as a practical way to improve the training signal in cooperative MARL.

Keywords: Multi-Agent Reinforcement Learning · Credit Assignment · Peer-Relative Utility Shaping · Cooperative MARL

1 Introduction

Multi-Agent Reinforcement Learning (MARL) has become a core framework for studying decentralized cooperation, because it captures how multiple agents learn coordinated behavior from shared outcomes and local observations. Its importance continues to grow in domains such as autonomous systems, embodied control, and other cooperative decision-making problems [1]. As these applications become more realistic, however, the main challenge is no longer just learning a high-return joint policy, but learning one whose training signal is sufficiently informative for each individual agent.

A standard paradigm for cooperative MARL is *Centralized Training with Decentralized Execution* (CTDE), in which a centralized learner can exploit global information during training while each agent executes from its own local history at test time. CTDE has become popular because it balances training efficiency and deployment practicality, but it also makes *Individual Credit Assignment* (ICA) especially important [2, 3]. The learner must decide how team-level return should influence each agent's local utility estimate, even though the observed outcome reflects delayed rewards, partial observability, and coupled decisions from the whole team [4]. When this attribution is weak or noisy, training can become slow, unstable, or misleading.

The difficulty is amplified by several properties of cooperative MARL. Joint action spaces grow exponentially with the number of agents, so directly reasoning over all interaction effects is expensive. Reward feedback is often sparse or delayed, making it hard to separate strategically important actions from incidental ones. Strong inter-agent dependence means that the value of one agent's action can change substantially depending on what its teammates do at the same step. These factors together make ICA not only a modeling problem, but also a supervision problem: the optimization signal itself may fail to expose which local behaviors should be reinforced [3, 4].

Existing ICA methods are often organized into two broad families. *Implicit* approaches, including VDN [5], QMIX [6], and related value-factorization methods, attribute credit through structural decomposition of the joint value. *Explicit* approaches, including COMA [7], MADDPG [8], and MAAC [9], attempt to estimate each agent's effect on team performance more directly [10, 11]. Both families are valuable, but they expose a recurring trade-off. Implicit methods are usually scalable and easy to optimize, yet their factorization form can hide fine-grained interaction structure. Explicit methods can be more interpretable, but they often introduce higher computational cost or stronger architectural dependence [12, 13]. This leaves room for lightweight auxiliary mechanisms that improve the learning signal without replacing the host backbone.

Motivated by this gap, we propose **GrCA-Decor**, a plug-and-play decorator for value-based CTDE methods. Its core ingredient is a *group-relative counterfactual advantage* (**GrCA**) that compares an agent's chosen action value with a same-action peer reference across agents. In our setting, GrCA should be interpreted as a peer-relative utility-shaping signal on the critic side rather than as a full replacement for conventional counterfactual action marginalization; throughout this paper, the term *counterfactual* is used in this narrower peer-relative sense.[1] The resulting relative signal is injected back into the per-agent utility before mixing, so the host algorithm receives additional training information while keeping its original optimization pipeline and decentralized execution interface.

We evaluate GrCA-Decor on the *StarCraft Multi-Agent Challenge* (SMAC) and *Google Research Football* benchmarks by integrating it into representative

[1] Here, *counterfactual* denotes a peer-relative contrast, not standard self-action marginalization or a causal marginal-contribution estimate.

value-based baselines such as VDN and QMIX. Across both benchmark families, the GrCA-enhanced variants generally converge more favorably and achieve stronger late-stage performance, with clearer gains on harder coordination settings where credit ambiguity is most severe.

In summary, the major contributions of this work include:

- We propose **GrCA-Decor**, a lightweight decorator that introduces peer-relative utility shaping as an auxiliary training signal for value-based CTDE algorithms.
- We formulate **GrCA** by constructing a same-action group reference and injecting the resulting relative correction into per-agent utilities before the host mixer, without changing decentralized execution.
- We demonstrate favorable gains on representative SMAC and Google Research Football tasks, and further analyze the method through weighting and variant ablations.

2 Related Work

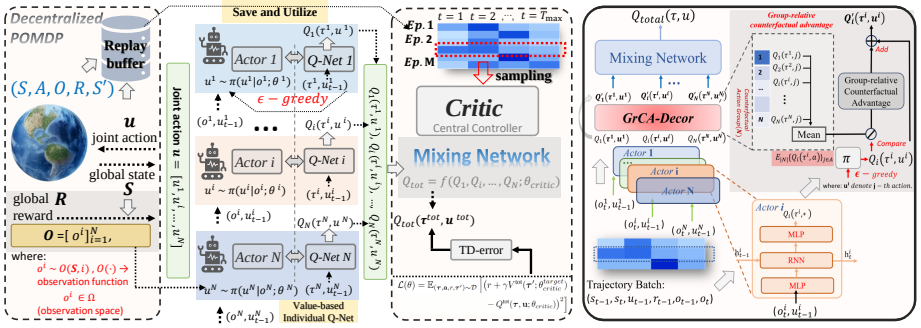
Cooperative MARL under CTDE remains challenging because agents must learn from shared rewards while acting under partial observability [2, 14]. The central question is how to attribute team-level success to individual decisions, namely the ICA problem [10, 11].

Existing ICA methods can be grouped into two broad categories [2, 10]. **Implicit methods** perform credit assignment through value factorization. Representative examples include VDN [5], which uses additive decomposition, QMIX [6], which introduces a state-conditioned monotonic mixing network, and Weighted QMIX [15], which further adjusts the mixing behavior for optimistic and pessimistic value estimation. **Explicit methods** attempt to model each agent's effect on the global value more directly. Examples include COMA [7], which uses an action-conditional baseline, MAAC [9], which models inter-agent dependence through attention, and Shapley-style credit assignment methods [10, 16].

These lines of work highlight a recurring trade-off. Implicit approaches are usually efficient and easy to optimize, but they can hide interaction structure inside the mixer. Explicit approaches are often more interpretable, but they can be computationally expensive or specialized to particular learning architectures [2, 3, 13]. Our goal is therefore more modest than replacing existing ICA formulations: we seek a lightweight auxiliary signal that can strengthen value-based CTDE training without requiring a new backbone.

3 Preliminaries

A cooperative multi-agent task is commonly modeled as a Decentralized POMDP [17], defined by the tuple $\mathcal{G} = \langle I, S, A, P, R, \Omega, O, N, \gamma \rangle$. Here, I is the set of N agents, S is the global state space, A is the per-agent action space, $P(s' \mid s, \mathbf{u})$



(a) Actor-Critic based methods under CTDE paradigm.

(b) Our *group-relative counterfactual advantage* decorator (**GrCA-Decor**) for credit assignment.

Fig. 1. Illustration of value-based CTDE credit assignment. Part (a) shows a standard host architecture, while Part (b) adds the proposed GrCA-Decor module, which injects a group-relative auxiliary signal into the per-agent utilities before mixing.

is the transition function, $R(s, \mathbf{u})$ is the shared reward, Ω is the observation space, $O(s, i)$ provides the local observation for agent i, and $\gamma \in [0, 1)$ is the discount factor. At each step, agent i receives an observation o_i, maintains an action-observation history τ_i, and selects an action through a decentralized policy $\pi_i(u_i^t \mid \tau_i, u_i^{t-1})$.

Under CTDE, training can use the global state while execution remains decentralized. The joint action-value function is

$$Q^{\text{tot}}(s, \mathbf{u}) = r(s, \mathbf{u}) + \gamma \mathbb{E}_{s'} \left[\max_{\mathbf{u}'} Q^{\text{tot}}(s', \mathbf{u}') \right]. \tag{1}$$

In practice, it is approximated by a neural critic $Q^{\text{tot}}(\boldsymbol{\tau}, \mathbf{u}; \theta)$ and optimized with the temporal-difference loss

$$\mathcal{L}(\theta) = \mathbb{E}_{(\boldsymbol{\tau}, \mathbf{u}, r, \boldsymbol{\tau}') \sim \mathcal{D}} \left[\left(r + \gamma V^{\text{tot}}(\boldsymbol{\tau}'; \theta^-) - Q^{\text{tot}}(\boldsymbol{\tau}, \mathbf{u}; \theta) \right)^2 \right], \tag{2}$$

where $\mathcal{D}$ is the replay buffer and θ^- denotes the target-network parameters. Individual action values $Q_i(\tau_i, u_i; \phi_i)$ must ultimately support this joint estimate through a mixing or decomposition mechanism, which is exactly where ICA becomes important.

4 Method

GrCA-Decor is a training-time decorator for value-based CTDE algorithms. As illustrated in Fig. 1, it augments each agent's utility with a group-relative auxiliary signal before the utilities are passed to the host mixer. At execution time, the deployed decentralized policy is unchanged: each agent still acts with its original local utility estimator Q_i. The decorator therefore affects learning, not the test-time interface.

Let $\tau = (\tau^1, \ldots, \tau^N)$ denote the agents' action-observation histories and $\mathbf{u} = (u^1, \ldots, u^N)$ the joint action. For agent i, the actor outputs per-action value estimates $Q_i(\tau^i, a)$ for all $a \in A$. We assume a shared discrete action vocabulary within each task and approximately comparable value scales across agents. When action masking is used, the group reference for action a_j is computed only over agents for which a_j is available.

Group-relative reference. For each action $a_j \in A$, we compute a same-action group baseline:

$$\bar{Q}^{(a_j)} = \frac{1}{|\mathcal{I}(a_j)|} \sum_{k \in \mathcal{I}(a_j)} Q_k(\tau^k, a_j), \tag{3}$$

where $\mathcal{I}(a_j)$ is the set of agents for which action a_j is valid at the current step. We use the term *group-relative counterfactual* in a narrow sense: the current action is compared against a same-action peer reference rather than against the agent's own marginalized alternative actions.

The resulting GrCA signal is

$$\mathrm{GrCA}_i(\tau^i, u^i) = Q_i(\tau^i, u^i) - \bar{Q}^{(u^i)}. \tag{4}$$

Intuitively, GrCA_i measures whether agent i evaluates its chosen action better or worse than the group-level reference for taking that same action. In our work, it functions as a critic-side relative correction signal rather than as a full substitute for action-level counterfactual baselines; accordingly, the term *counterfactual* should be read in this narrow peer-relative sense. The value should not be interpreted as a direct causal or marginal contribution estimate.

Enhanced training utilities. We inject this relative signal back into the utility used by the mixer during training:

$$Q'_i(\tau^i, u^i) = Q_i(\tau^i, u^i) + \lambda\, \mathrm{GrCA}_i(\tau^i, u^i), \tag{5}$$

where λ is a weighting coefficient. Equivalently, this is a linear peer-centering transformation, $Q'_i = (1+\lambda)Q_i - \lambda\bar{Q}^{(u^i)}$, so the contribution of GrCA is best understood as a simple utility-shaping heuristic rather than a new causal credit estimator. In all main experiments, we use $\lambda = 1$ unless otherwise noted. The enhanced utilities are then aggregated by the host mixer:

$$Q_{\mathrm{tot}}(\tau, \mathbf{u}) = \mathrm{Critic}\big(Q'_1(\tau^1, u^1), \ldots, Q'_N(\tau^N, u^N); \psi\big)\,, \tag{6}$$

where ψ denotes the critic parameters. Because execution still uses the original local utilities Q_i, the decentralized action-selection rule of the host backbone is preserved.

Relation to decentralized greedy consistency. Since Q'_i contains a peer reference, the standard Individual-Global-Max (IGM) argument for monotonic value factorization does not automatically transfer from Q_i to Q'_i. It is preserved under the sufficient action-ranking condition

$$\arg\max_a Q_i(\tau^i, a) = \arg\max_a \Big[Q_i(\tau^i, a) + \lambda\left(Q_i(\tau^i, a) - \bar{Q}^{(a)}\right)\Big]\,,$$

for every agent at the evaluated history. Otherwise, GrCA-Decor should be viewed as critic-side utility shaping rather than as a separate execution policy with a general IGM guarantee.

Training with the decorator. The target critic uses the GrCA-enhanced utilities at the next step:

$$y = r + \gamma Q^{-}_{\mathrm{tot}}(\boldsymbol{\tau}', \boldsymbol{u}^{*'}), \tag{7}$$

where $u_i^{*'} = \arg\max_{u^i} Q_i(\tau'^i, u^i)$ remains the host backbone's decentralized greedy action. It is execution-consistent, but not necessarily the maximizer of the enhanced joint value. The temporal-difference error is $\delta = Q_{\mathrm{tot}}(\boldsymbol{\tau}, \boldsymbol{u}) - y$, and the optimization objective is unchanged: $\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}}[\delta^2]$. This training-time design should be understood as utility shaping rather than as a new execution-time action rule: the decorator changes the learning signal seen by the host backbone, while the deployed policy still acts through the original decentralized utilities Q_i. The resulting train-test asymmetry is limited to the critic-side supervision pathway, because both greedy action selection and decentralized execution still rely on the original local utilities. By reshaping only the mixer-side temporal-difference supervision, the decorator can influence how the local utilities are trained without redefining the decentralized action rule itself.

In this sense, GrCA-Decor is closer to a structured critic regularizer or utility-shaping mechanism than to a change in the deployed policy class. The method introduces no additional trainable blocks, and λ is a scalar hyperparameter. Its structural overhead is a simple group reduction over agent-action values, i.e., linear in the number of valid utility entries at each step. In principle, the same interface can be applied to value-based CTDE architectures that aggregate per-agent utilities. In our work, the full learning-curve evaluation focuses on VDN [5] and QMIX [6]. We additionally include a compact late-stage transfer check on QTRAN [18] and QPLEX [19] in Sec. 5.4 to illustrate compatibility.

5 Experiments

5.1 Experimental Settings

We evaluate GrCA-Decor primarily on the *StarCraft Multi-Agent Challenge* (SMAC), a standard benchmark for cooperative MARL. Fig. 2 illustrates representative scenarios used in the SMAC experiment groups; these labels are coarse experimental groupings rather than a formal difficulty taxonomy. We additionally test on *Google Research Football* (Gfootball)[2], as depicted in Fig. 3. The representative Gfootball tasks are *11_vs_11_easy_stochastic*, *11_vs_11_hard_stochastic*, *academy_3_vs_1_with_keeper*, and *academy_5_vs_5*.

We instantiate the decorator on two value-based baselines, VDN [5] and QMIX [6], yielding VDN-GrCA and QMIX-GrCA. For reference, we also report COMA and QPLEX as additional baselines, and later provide a compact

[2] Official project page: `https://github.com/google-research/football`.

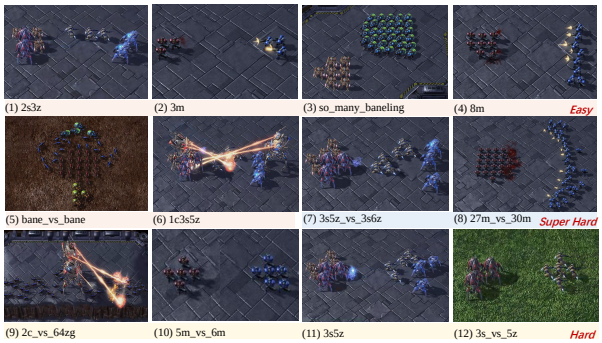


Fig. 2. Representative SMAC scenarios used in our experiments.

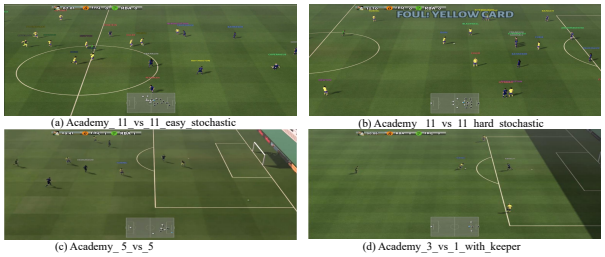


Fig. 3. Representative Gfootball scenarios used for evaluation.

transfer check on QTRAN/QPLEX with the same decorator. Weighted QMIX is discussed as a related factorization method, but is not included in the present experimental suite. Unless explicitly changed by the decorator, we inherit the optimizer, exploration schedule, replay configuration, and target-network update rules from the adopted public implementations. Each checkpoint is evaluated every 10,000 environment steps using 32 test episodes.

For reproducibility, we use publicly available MARL implementations[3]. Unless otherwise stated, all backbone hyperparameters are kept unchanged so that the comparison isolates the effect of the proposed decorator. We report the main SMAC comparison as learning curves and summarize the main Gfootball comparison with late-stage win rates; the Gfootball curves in Fig. 9 are used for the variant ablation. In each table, late-stage values are computed over a fixed converged-stage evaluation window and should be read as descriptive point summaries when no interval is shown. The Gfootball ablation curves are averaged over five independent runs, with shaded bands indicating variability. We report computational cost at the level of structural overhead: GrCA adds no trainable blocks and only a lightweight reduction over agent-action values, while wall-clock profiling is left for future work.

[3] We build on the public MARL codebase at `https://github.com/starry-sky6688/MARL-Algorithms`.

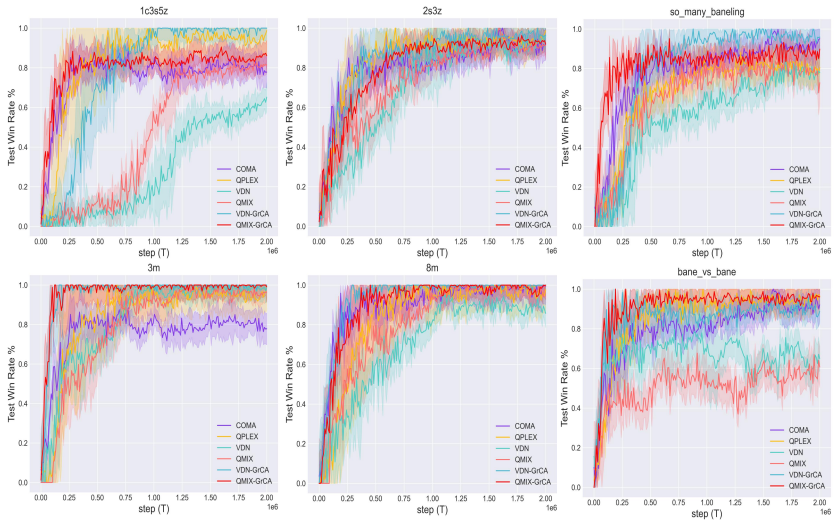


Fig. 4. Performance comparison on representative easy SMAC maps.

5.2 Main Results on SMAC

Results on easy maps: As shown in Fig. 4, the GrCA-enhanced variants generally converge faster and reach comparable or better win rates than their host backbones on easy SMAC scenarios. The gains are especially visible on maps such as `3m`, `8m`, and `1c3s5z`, suggesting that even in relatively simple settings the extra group-relative signal can make credit propagation easier for the mixer.

Results on hard maps: On the hard maps in Fig. 5, the relative advantage of the GrCA-enhanced variants becomes more pronounced, particularly on `2c_vs_64zg` and `5m_vs_6m`. These trends are consistent with the view that the additional training signal is more helpful when coordination becomes longer-horizon and more ambiguous.

Results on super-hard maps: On the super-hard maps shown in Fig. 6, all methods remain challenged, but the GrCA variants still tend to learn more stably and finish above their corresponding host baselines. The gains are especially noticeable for VDN, indicating that simple value-factorization backbones may benefit the most from the extra relative shaping signal.

5.3 Main Results on Gfootball

We further evaluate GrCA-Decor on Gfootball, which requires fine-grained coordination under partial observability. Table 1 shows that both VDN-GrCA and QMIX-GrCA improve over their respective host backbones on the evaluated tasks. The absolute gains are larger on some scenarios than on others, but the overall trend remains favorable, especially for QMIX-GrCA on `3_vs_1_keeper` and `11_vs_11_easy`. We therefore view the Gfootball results as complementary evidence that the decorator can transfer beyond SMAC. As above, the table

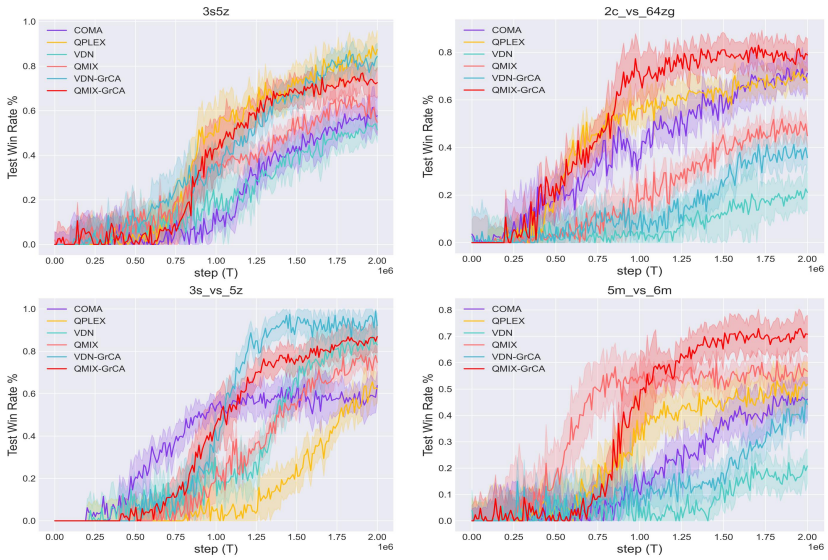


Fig. 5. Performance Comparison on several '*hard*' SMAC maps, i.e., `3s5z`, `2c_vs_64zg`, `5m_vs_6m` and `3s_vs_5z`.

Table 1. Descriptive late-stage test win-rate summary (%) on Gfootball.

Method	11_vs_11_easy	11_vs_11_hard	3_vs_1_keeper	5_vs_5
COMA	78.2	58.6	69.4	66.8
VDN	80.5	60.3	71.8	69.2
QMIX	82.1	62.7	74.6	71.5
QPLEX	83.4	63.9	76.2	73.1
VDN-GrCA	84.1	64.0	77.3	71.6
QMIX-GrCA	**85.3**	**64.8**	**78.1**	**74.2**

should be read as a converged-stage descriptive summary rather than as a replacement for full statistical testing.

5.4 Extension to QTRAN and QPLEX

To further examine the generality of the proposed decorator, we include an additional evaluation on QTRAN and QPLEX, two stronger value-factorization backbones. Table 2 reports late-stage test win rates on five representative scenarios, including two Gfootball tasks and three SMAC maps. For a controlled comparison, the same implementation pipeline is used, the backbone-specific hyperparameters are kept unchanged, and the default GrCA setting is adopted throughout. This experiment is reported as a compact descriptive summary rather than a full learning-curve study, and should therefore be viewed as supplementary evidence on cross-backbone applicability.

Two observations are noteworthy. First, QPLEX is consistently stronger than QTRAN on all selected scenarios, which matches the higher expressive power

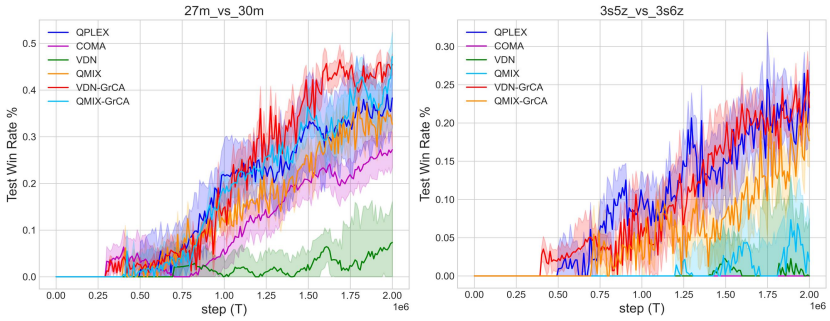


Fig. 6. Comparison on representative super-hard SMAC maps.

Table 2. Descriptive late-stage test win rates (%) of QTRAN/QPLEX and their GrCA-enhanced variants on representative Gfootball and SMAC tasks. Gain annotations in the GrCA columns are computed with respect to the corresponding host backbone.

Benchmark	Scenario	QTRAN	QTRAN-GrCA	QPLEX	QPLEX-GrCA
Gfootball	11_vs_11_easy	80.7	82.6 +2.4%↑	83.4	**85.1** +2.0%↑
	11_vs_11_hard	61.2	67.4 +10.1%↑	63.9	**65.7** +2.8%↑
SMAC	3s5z	71.6	74.0 +3.4%↑	76.1	**78.4** +3.0%↑
	5m_vs_6m	87.9	89.0 +1.3%↑	91.1	**91.9** +0.9%↑
	2c_vs_64zg	65.8	72.3 +9.9%↑	70.9	**73.6** +3.8%↑

of the duplex dueling factorization. Second, GrCA improves both backbones on Gfootball and SMAC, with relative gains of 1.3%–10.1% for QTRAN and 0.9%–3.8% for QPLEX in this table. The gains are more pronounced on the harder SMAC maps and on `11_vs_11_hard`, while they become smaller on the relatively saturated `5m_vs_6m` case. Overall, these results suggest that GrCA can provide a complementary training signal for several value-based CTDE architectures.

These results should be interpreted with appropriate caution. No additional large-scale hyperparameter sweep was conducted specifically for QTRAN or QPLEX, and the table is not intended to serve as a definitive benchmark among advanced mixers. Nevertheless, the favorable trend across the selected tasks provides supporting evidence that the proposed decorator is compatible with a broader family of value-based CTDE methods and is not restricted to VDN and QMIX.

5.5 Episode-Level Diagnostic View of GrCA

Beyond benchmark win rates, we clarify how the proposed signal can be inspected at the agent level. Specifically, we define an episode-level diagnostic

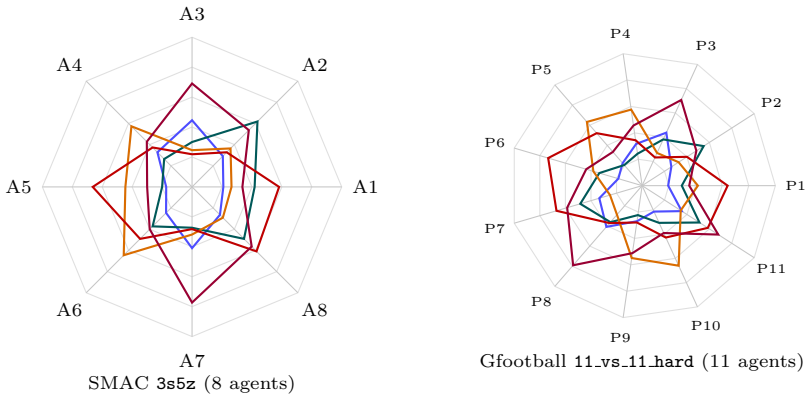


Fig. 7. Illustrative episode-level diagnostic visualization of normalized GrCA magnitudes at five representative decision times (t_1 to t_5). Each spoke denotes one agent.

magnitude

$$d_i^{(t)} = \frac{|\mathrm{GrCA}_i^{(t)}|}{\max_k |\mathrm{GrCA}_k^{(t)}| + \epsilon}, \tag{8}$$

where $d_i^{(t)} \in [0, 1]$ denotes the normalized strength of agent i's GrCA signal at decision time t. This quantity is not used for training; it is only intended as a qualitative diagnostic view of how the relative correction is distributed across agents during one episode.

Fig. 7 illustrates this diagnostic protocol for one representative SMAC task and one Gfootball task. The intended use is not to claim a new benchmark result from these plots, but to show how GrCA can be inspected as a time-varying agent-level signal. Because each profile is rescaled to the same total mass, the plots show only relative redistribution across agents. They should therefore be understood as a qualitative first step; future diagnostics should compare GrCA against quantitative marginal-contribution or intervention-based measures.

5.6 Ablation Studies

Ablation on the weighting coefficient λ. We first study the effect of the weighting coefficient λ in QMIX-GrCA on three representative SMAC maps. For compact comparison, Table 3 summarizes the converged-stage performance of each run using a fixed late-stage evaluation window rather than a single peak checkpoint.

The results indicate that moderate values of λ, especially 0.5 and 1, provide the most reliable behavior across maps. Very small weights underuse the auxiliary signal, while very large weights tend to over-amplify the relative correction. Since $\lambda = 1$ performs strongly and consistently, we use it as the default configuration in the rest of the paper.

Ablation on GrCA variants. Our default GrCA uses a simple same-action group mean as the reference baseline. To test whether this choice matters, we compare it with three alternative variants:

Table 3. Effect of the GrCA weighting coefficient λ on late-stage SMAC performance.

Maps/GrCA Weight	λ=0.1	λ=0.2	λ=0.5	λ=1	λ=2	λ=5
2s3z	0.89	0.91	**0.93**	0.92	0.84	0.72
bane_vs_bane	0.84	0.89	0.93	**0.95**	0.88	0.76
3s5z	0.66	0.73	0.73	**0.76**	0.74	0.58

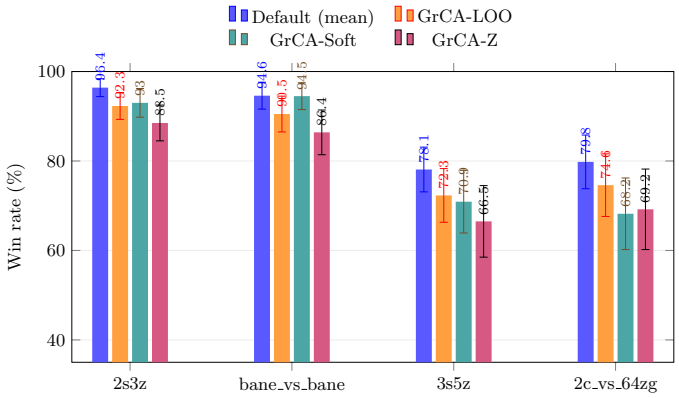


Fig. 8. Ablation on alternative GrCA variants on challenging SMAC maps. The bars summarize late-stage performance, and the error bars indicate variability across five runs.

- **GrCA-LOO (leave-one-out mean).** We compare agent i against the mean Q-value of the remaining agents for the same action. $\mathrm{GrCA}_i^{\mathrm{LOO}} = Q_i(\tau^i, u^i) - 1/(|N|-1)\sum_{k\neq i} Q_k(\tau^k, u^i)$.
- **GrCA-Soft (softmax-weighted mean).** For this variant, we replace the uniform mean by a softmax-weighted mean to emphasize stronger counterfactual peers:

$$\mathrm{GrCA}_i^{\mathrm{Soft}} = Q_i(\tau^i, u^i) - \sum_{k=1}^{|N|} w_k(u^i)\, Q_k(\tau^k, u^i), \tag{9}$$

where

$$w_k(u^i) = \frac{\exp(Q_k(\tau^k, u^i)/\tau)}{\sum_j \exp(Q_j(\tau^j, u^i)/\tau)}, \quad \tau = 1. \tag{10}$$

- **GrCA-Z (z-normalized advantage).** We normalize the advantage by the group standard deviation for the same action:

$$\mathrm{GrCA}_i^{\mathrm{Z}} = \frac{Q_i(\tau^i, u^i) - \mu(u^i)}{\sigma(u^i) + \epsilon}, \tag{11}$$

where

$$\mu(u^i) = \frac{1}{N}\sum_k Q_k(\tau^k, u^i), \quad \sigma(u^i) = \sqrt{\frac{1}{N}\sum_k (Q_k(\tau^k, u^i) - \mu(u^i))^2}, \tag{12}$$

and $\epsilon = 10^{-5}$.

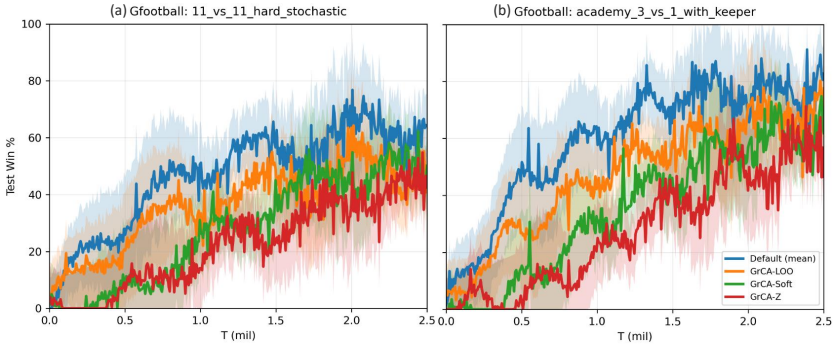


Fig. 9. Ablation comparison of GrCA variants on Gfootball.

We plug each variant into QMIX-GrCA with the same training setup and $\lambda = 1$. On SMAC, we summarize late-stage performance on four representative maps. On Gfootball, we additionally compare the learning curves of the variants on `11_vs_11_hard_stochastic` and `3_vs_1_with_keeper`, where the curves are averaged over five independent runs with shaded variability bands.

On SMAC (Fig. 8), the default mean-based variant achieves the strongest or most stable late-stage performance across the selected maps. The alternative variants remain competitive on some easier cases, but their behavior is generally less reliable on the harder tasks. On Gfootball (Fig. 9), we can observe a similar trend: the default variant converges faster and ends at a higher level on both tasks. Taken together, these ablations suggest that a simple uniform group mean is a strong default choice for constructing the GrCA signal.

6 Discussion

GrCA-Decor is positioned differently from many existing credit-assignment methods. Rather than replacing value factorization itself, as in VDN and QMIX-style designs [5, 6], or introducing heavier explicit interaction modeling as in COMA and MAAC [7, 9], it adds a lightweight peer-relative shaping signal to the existing CTDE pipeline. The proposed signal does not require a new mixer, additional trainable modules, or costly combinatorial reasoning over joint actions, and it is intended to provide a more structured training cue than using raw per-agent utilities alone. In that sense, GrCA-Decor is closer to an auxiliary utility-shaping cue than to a new backbone, complementing both factorization-based and explicit credit-assignment methods [10, 11]. Overall, these suggest that MARL training signals are improved not only by making critics more expressive, but also by making supervision more relational.

A practical limitation is that the same-action peer mean assumes aligned action semantics and roughly comparable utility scales; strongly heterogeneous roles, non-shared parameterization, or agent-specific shifts and rescalings require further sensitivity checks. Lightweight peer-relative signals of this kind could

therefore be a useful direction for future CTDE research, especially in harder cooperative settings where attribution ambiguity is most severe [4].

7 Conclusion

We presented GrCA-Decor, a lightweight decorator for cooperative MARL that adds a peer-relative utility-shaping signal to value-based CTDE training. The method compares each agent's chosen action value with a same-action group reference, then injects the resulting relative signal back into the individual Q-value before mixing. This design preserves the deployed decentralized action rule of the host algorithm while changing the critic-side supervision used during training. Across SMAC and Gfootball benchmarks, GrCA-Decor often improves the tested host baselines and remains competitive under different weighting choices and GrCA variants. Because the method is modular and structurally lightweight, it can be integrated into value-based MARL pipelines with limited engineering overhead. In future work, we will broaden the empirical study, add wall-clock or throughput measurements, quantify original-versus-enhanced greedy mismatch, compare target-action constructions, and test more value-based backbones.

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References

1. Zhang, R., Hou, J., Walter, F., Gu, S., Guan, J., Röhrbein, F., Du, Y., Cai, P., Chen, G., Knoll, A.: Multi-agent reinforcement learning for autonomous driving: A survey (2024), https://arxiv.org/abs/2408.09675
2. Orr, J., Dutta, A.: Multi-agent deep reinforcement learning for multi-robot applications: A survey. Sensors **23**(7) (2023). https://doi.org/10.3390/s23073625
3. Ning, Z., Xie, L.: A survey on multi-agent reinforcement learning and its application. Journal of Automation and Intelligence **3**(2), 73–91 (2024). https://doi.org/10.1016/j.jai.2024.02.003
4. Du, W., Ding, S.: A survey on multi-agent deep reinforcement learning: from the perspective of challenges and applications. Artificial Intelligence Review **54**(5), 3215–3238 (2021). https://doi.org/10.1007/s10462-020-09938-y
5. Sunehag, P., Lever, G., Gruslys, A., Czarnecki, W.M., Zambaldi, V., Jaderberg, M., Lanctot, M., Sonnerat, N., Leibo, J.Z., Tuyls, K., Graepel, T.: Value-decomposition networks for cooperative multi-agent learning based on team reward. In: Proceedings of the 17th International Conference on Autonomous Agents and MultiAgent Systems. pp. 2085–2087. AAMAS '18, International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC (2018)
6. Rashid, T., Samvelyan, M., Schroeder, C., Farquhar, G., Foerster, J., Whiteson, S.: QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning. In: Dy, J., Krause, A. (eds.) Proceedings of the 35th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 80,

pp. 4295–4304. PMLR (10–15 Jul 2018), https://proceedings.mlr.press/v80/rashid18a.html
7. Foerster, J.N., Farquhar, G., Afouras, T., Nardelli, N., Whiteson, S.: Counterfactual multi-agent policy gradients. In: Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence and Thirtieth Innovative Applications of Artificial Intelligence Conference and Eighth AAAI Symposium on Educational Advances in Artificial Intelligence. AAAI'18/IAAI'18/EAAI'18, AAAI Press (2018)
8. Lowe, R., Wu, Y., Tamar, A., Harb, J., Abbeel, P., Mordatch, I.: Multi-agent actor-critic for mixed cooperative-competitive environments. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. NIPS'17, Curran Associates Inc., Red Hook, NY, USA (2017)
9. Iqbal, S., Sha, F.: Actor-attention-critic for multi-agent reinforcement learning. In: Chaudhuri, K., Salakhutdinov, R. (eds.) Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 2961–2970. PMLR (09–15 Jun 2019), https://proceedings.mlr.press/v97/iqbal19a.html
10. Li, J., Kuang, K., Wang, B., Liu, F., Chen, L., Wu, F., Xiao, J.: Shapley counterfactual credits for multi-agent reinforcement learning. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. pp. 934–942. KDD '21, Association for Computing Machinery, New York, NY, USA (2021). https://doi.org/10.1145/3447548.3467420
11. Seo, M., Vecchietti, L.F., Lee, S., Har, D.: Rewards prediction-based credit assignment for reinforcement learning with sparse binary rewards. IEEE Access **7**, 118776–118791 (2019). https://doi.org/10.1109/ACCESS.2019.2936863
12. Jin, W., Tian, X., Wang, N., Wu, B., Shi, B., Zhao, B., Yang, G.: Representation-driven sampling and adaptive policy resetting for improving multi-agent reinforcement learning. Neural Networks **192**, 107875 (2025). https://doi.org/10.1016/j.neunet.2025.107875
13. Jin, W., Du, H., Zhao, B., Tian, X., Shi, B., Yang, G.: A comprehensive survey on multi-agent cooperative decision-making: Scenarios, approaches, challenges and perspectives (2025), https://arxiv.org/abs/2503.13415
14. Gronauer, S., Diepold, K.: Multi-agent deep reinforcement learning: a survey. Artificial Intelligence Review **55**(2), 895–943 (2022). https://doi.org/10.1007/s10462-021-09996-w, https://doi.org/10.1007/s10462-021-09996-w
15. Rashid, T., Farquhar, G., Peng, B., Whiteson, S.: Weighted qmix: expanding monotonic value function factorisation for deep multi-agent reinforcement learning. In: Proceedings of the 34th International Conference on Neural Information Processing Systems. NIPS '20, Curran Associates Inc., Red Hook, NY, USA (2020)
16. Wang, J., Zhang, Y., Kim, T.K., Gu, Y.: Shapley q-value: A local reward approach to solve global reward games. Proceedings of the AAAI Conference on Artificial Intelligence **34**(05), 7285–7292 (Apr 2020). https://doi.org/10.1609/aaai.v34i05.6220
17. Lauri, M., Hsu, D., Pajarinen, J.: Partially observable markov decision processes in robotics: A survey. IEEE Transactions on Robotics **39**(1), 21–40 (2023)
18. Son, K., Kim, D., Kang, W.J., Hostallero, D.E., Yi, Y.: QTRAN: Learning to factorize with transformation for cooperative multi-agent reinforcement learning. In: Chaudhuri, K., Salakhutdinov, R. (eds.) Proceedings of the 36th International Conference on Machine Learning. vol. 97, pp. 5887–5896. PMLR (09–15 Jun 2019)
19. Wang, J., Ren, Z., Liu, T., Yu, Y., Zhang, C.: Qplex: Duplex dueling multi-agent q-learning. In: 9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021. OpenReview.net (2021), https://openreview.net/forum?id=Rcmk0xxIQV